

IDENTIFYING SOUTH AFRICAN TOURISM TIME SERIES STRUCTURE USING SINGULAR SPECTRUM ANALYSIS

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Summary: Singular spectrum analysis is a powerful non-parametric time series method that applies singular value decomposition to a Hankel structured matrix. The method can handle complex time series structures that include combinations of polynomials, sinusoids and exponentials. The effectiveness of the method is illustrated by application to a South African tourism time series.

1. Introduction

Singular Spectrum Analysis (SSA) is a powerful non-parametric time series technique that found its origins in the field of Physics (Takens, 1981; Broomhead and King, 1986). Golyandina, Nekrutkin and Zhigljavsky (2001) provide a thorough introduction to SSA and can be consulted in gaining insight into the underlying theory and applications.

According to SSA methodology a time series $\{y_t\}_{t=1}^N$ is unfolded into the column vectors of a Hankel structured matrix

$$\mathbf{X}_{L \times (N-L+1)} = (x_{ij})_{i,j=1}^{L, N-L+1} = \begin{bmatrix} y_1 & y_2 & \cdots & y_{N-L+1} \\ y_2 & y_3 & \cdots & y_{N-L+2} \\ y_3 & y_4 & \ddots & y_{N-L+3} \\ \vdots & \vdots & \ddots & \vdots \\ y_L & y_{L+1} & \cdots & y_N \end{bmatrix}. \quad (1)$$

Note that anti-diagonal elements of secondary diagonals in this matrix are equal. The matrix has been coined the *trajectory matrix* in SSA literature and places a one-dimensional time series into a multivariate framework. The dimension L into which the column vectors are unfolded is called the *window length* and restricted by the choice $2 \leq L \leq \text{floor}[(N+1)/2]$.

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Buchstaber (1994) showed that time series sampled from the following broad class of functions with an additive property can be handled by SSA, viz.

$$y(t) = \sum_{k=1}^K p_k(t) \exp(\alpha_k t) \sin(2\pi\omega_k t + \phi_k) \quad (2)$$

where $p_k(t)$ indicate polynomials. Golyandina et al. (2001) indicated that SSA can actually handle a much broader class of functions than the above in the form of finite difference equations or so-called linear recurrent formulae (LRF) of the form

$$y_{t+r} = \sum_{k=1}^r a_k y_{t+r-k}, \quad 1 \leq t \leq N-r \quad (3)$$

where $a_1, \dots, a_r$ are coefficients and r is the dimension (structure) of the time series. It is clear that SSA can handle a wide variety of time series structure which can include trend with/without seasonality. The interested reader can refer to Golyandina *et al.* (2001) to further understand how singular value decomposition is an optimal method used to extract signal structures from an observed time series.

2. Basic SSA Methodology

Golyandina et al. (2001) proposes four stages that are applied during what the authors coined “Basic SSA”, viz. embedding, singular value decomposition, grouping and diagonal averaging. The authors also introduce an additional concept which they call “separability” of elementary SSA components, which can essentially assist with the grouping stage in SSA and general decomposition of time series structure.

2.1. Stages in SSA

2.1.1. Embedding

The embedding stage in SSA essentially entails the unfolding of an observed time series into the column vectors of a Hankel structured matrix, called the trajectory matrix. During this stage a one-dimensional time series $\{y_t\}_{t=1}^N$ is unfolded into the column vectors of a Hankel structured matrix where the column vectors are formed such that $\mathbf{x}_i = (y_i, \dots, y_{i+L-1})^T \in \mathbf{R}^L$ for $i = 1, \dots, N-L+1$ and L is the dimension of the Euclidean space into which the column vectors are mapped. This places a one-dimensional time series into a multivariate framework. The dimension L of the Euclidean space into which the column vectors are folded is called the embedding dimension (Physics literature) or window length (later SSA literature). The window length is governed by the choice $2 \leq L \leq \text{floor}[(N+1)/2]$. By application of this unfolding method it is clear that anti-diagonal elements of secondary diagonals in this matrix are equal and gives rise to the Hankel structure of the matrix.

2.1.2. Singular Value Decomposition (SVD)

The second stage in SSA entails the use of singular value decomposition (SVD) to extract the one-dimensional “structure” components from the Hankel structured trajectory matrix. According to Golyandina et al. (2001) the purpose is to form elementary matrices $(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_d)$ that are each

of rank one and that the sum of these elementary matrices would produce the trajectory matrix, i.e. $\mathbf{X}_{L \times (N-L+1)} = (\mathbf{X}_1 + \mathbf{X}_2 + \dots + \mathbf{X}_d)$. This is actually the spectral decomposition of a matrix, i.e. $\mathbf{X}_{L \times (N-L+1)} = \sum_{i=1}^d \mathbf{X}_i = \sum_{i=1}^d \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T$, where SSA derives the nomenclature from. Note that any given elementary matrix can be reconstructed using $\mathbf{X}_i = \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T$ for $i = 1, \dots, d$. Here d denotes the rank of the trajectory matrix. Eigenvalues are extracted from the non-centered square matrix $\mathbf{X}\mathbf{X}^T$, which results in eigentriples $(\lambda_i, \mathbf{u}_i, \mathbf{v}_i)$ for $i = 1, \dots, L$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_L > 0$ are the ordered (in decreasing order of magnitude) eigenvalues and $(\mathbf{u}_i, \mathbf{v}_i)$ the orthonormal singular vectors. Note that $\mathbf{u}_i^T \mathbf{u}_j = 0$ for $i \neq j$ since the singular vectors are mutually orthogonal and the same holds true that $\mathbf{v}_i^T \mathbf{v}_j = 0$ for $i \neq j$. The ratio $\lambda_i / \sum_{i=1}^d \lambda_i$ can be used to deduce the importance of each elementary matrix's contribution to the overall structure in the trajectory matrix. In this regard the importance of the structure explained by the r leading singular vector can be calculated using $\sum_{i=1}^r \lambda_i / \sum_{i=1}^d \lambda_i$. The latter, multiplied by 100, is customarily coined in classical multivariate statistical literature as the percentage variation explained by the r leading singular vectors. The interested reader can consult Section 4 in Golyandina et al. (2001) for theoretical aspects which deal with the optimality of using SVD in the context of SSA.

It is important to understand how structure in time series can contribute to the rank of the trajectory matrix. Buchstaber (1994) is the earliest reference that succinctly elucidates the manner in which rank is contributed to the trajectory matrix due to the additive property of certain mathematical functions' unfolding. Buchstaber (1994) noted how functions with an additive property would contribute rank to the trajectory matrix and this notion was expanded later by Golyandina et al. (2001) to the much wider linear recurrent formulae (LRF) concept, which actually entails a very broad class of linear difference equations that SSA can handle. Table 1, below, summarizes the rank of the trajectory matrices formed using known mathematical functions which SSA can handle as presented by Buchstaber (1994) and derivations can be found in De Klerk (2002).

Table 1: Ranks of Hankel structured trajectory matrices formed from known mathematical functions.

Function	Rank (d) of Hankel structured trajectory matrix
$y_t = \text{constant}, t = 1, \dots, N$	1
$y_t = a^t, t = 1, \dots, N$	1
$y_t = b_0 + b_1 t + b_2 t^2 + \dots + b_k t^k, t = 1, \dots, N$	$k + 1$
$y_t = \sin(2\pi\omega t + \phi), t = 1, \dots, N$	2 if $0 < \omega < \frac{1}{2}$; 1 if $\omega = \frac{1}{2}$

It is clear from the results in Table 1, above, that pairs of singular vectors would be used to explain the structure of sinusoids (seasonality) in observed time series when $0 < \omega < \frac{1}{2}$. A single singular vector would be used in the SSA context to model exponential growth trend and that linear trend would be modeled using a pair of singular vectors. In certain cases so-called slow varying trend would also be modeled by individual singular vectors (Golyandina et al., 2001).

2.1.3. Grouping

As was clear from results in Table 1, above, some types of time series structure (notably higher order trend and seasonality) requires more than one singular vector to explain such time series structure. A typical monthly seasonal periodicity, for example, in an observed time series would require a pair of singular vectors to “model” seasonal time series structure. This notion leads to the concept of having to “group” singular vectors, by referring to their indices, in order to explain certain types of time series structure using basic SSA methodology.

The grouping stage, as proposed by Golyandina et al. (2001), forms groups of elementary matrices, where a group of indices of the original singular vectors, extracted using SVD of the trajectory matrix, is formed to represent some time series structure. The grouping stage denotes the indices of such grouping by the set $I = \{i_1, i_2, \dots, i_p\}$, where $i_1, i_2, \dots, i_p$ denote groups of indices. The matrix $\mathbf{X}_I$ corresponds to the group I as defined by $\mathbf{X}_I = \mathbf{X}_{i_1} + \dots + \mathbf{X}_{i_p}$ which in turn explains some time series structure. The original indices of the singular vectors extracted through a process of SVD, i.e. $J = 1, \dots, d$, are then split into disjoint subsets $I_1, \dots, I_m$ such that the trajectory matrix can then be written as $\mathbf{X} = \mathbf{X}_{I_1} + \dots + \mathbf{X}_{I_m}$. The latter can loosely be described as time series structures being modeled by that number of grouped matrices.

The grouping stage is notably important if the researcher is interested in the specific structures that a time series may exhibit. It is not difficult to show (De Klerk, 2002) that, in the case of a noise-free time series $\{y_t\}_{t=1}^N = \{\sin(2\pi\omega t + \phi), \sin(2\pi\omega(t + \Delta t) + \phi), \dots, \sin(2\pi\omega(t + (N-1)\Delta t) + \phi)\}$ where ω is the frequency, ϕ the phase angle and Δt sampling interval of the time series, that the column vectors of the trajectory matrix formed using this time series form linear combinations of the following independent vectors, viz.

$$\begin{aligned} \mathbf{y}_1 &= \begin{bmatrix} \cos(2\pi\omega t) & \cos(2\pi\omega t + \Delta t 2\pi\omega) & \dots & \cos(2\pi\omega t + (L-1)\Delta t 2\pi\omega) \end{bmatrix}^T \\ \mathbf{y}_2 &= \begin{bmatrix} \sin(2\pi\omega t) & \sin(2\pi\omega t + \Delta t 2\pi\omega) & \dots & \sin(2\pi\omega t + (L-1)\Delta t 2\pi\omega) \end{bmatrix}^T. \end{aligned} \quad (4)$$

Normalized vectors can be formed from the above using $\mathbf{u}_1 = \mathbf{y}_1 / \|\mathbf{y}_1\|$ and $\mathbf{u}_2 = \mathbf{y}_2 / \|\mathbf{y}_2\|$. It is clear from the above that a harmonic time series structure will be modeled in SSA using a pair of singular vectors. Note that the periodicity ($1/\omega$) in a time series is naturally preserved in the singular vector pairing. Any periodicity in the time series can therefore be ascertained by construction of periodograms based on the singular vectors.

Golyandina et al. (2001) recommends that scatterplots (cf. Figure 1, below) of singular vectors be formed to identify time series oscillatory structures. Figure 1, below, illustrates 6 possible harmonic

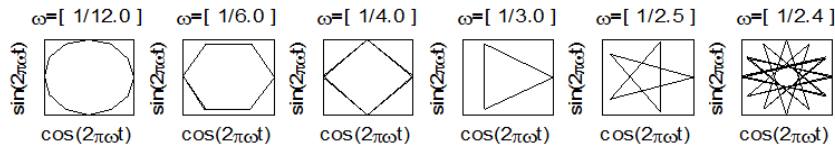


Figure 1: Scatterplots of singular vectors which represent oscillatory time series structures.

time series structures that may be present in a time series. Note how the number of vertices in each of the polygons correspond with the periodicity ($1/\varpi$) of the harmonic when the period is an integer.

2.1.4. Diagonal averaging

After the grouping stage, diagonal averaging of each grouped matrix ($\mathbf{X}_{I_1}, \dots, \mathbf{X}_{I_m}$) is required to yield Hankel structured matrices ($\tilde{\mathbf{X}}_{I_1}, \dots, \tilde{\mathbf{X}}_{I_m}$) and effectively the time series structures $\{\tilde{y}_t^{(1)}\}_{t=1}^N, \dots, \{\tilde{y}_t^{(m)}\}_{t=1}^N$. Golyandina *et al.* (2001) denotes this Hankelization operation by H . The Hankelization operation is applied to each matrix formed during the grouping stage to reconstruct a Hankel structured matrix which, when added together, will reconstruct the trajectory matrix. Hence, $\mathbf{X} = \tilde{\mathbf{X}}_{I_1} + \tilde{\mathbf{X}}_{I_2} + \dots + \tilde{\mathbf{X}}_{I_m}$ where $\tilde{\mathbf{X}}_{I_i} = H\mathbf{X}_{I_i}$. This Hankelization operation is also used to produce the signal components of the time series, viz.

$$\tilde{y}_t^{(k)} = \begin{cases} \frac{1}{s} \sum_{i=1}^s \tilde{x}_{t,s-i+1}^{(k)} & \text{for } 1 \leq s \leq L \\ \frac{1}{L} \sum_{i=1}^L \tilde{x}_{t,s-i+1}^{(k)} & \text{for } L < s \leq N-L+1 \\ \frac{1}{N-s+1} \sum_{i=1}^{N-s+1} \tilde{x}_{t,s-i+1}^{(k)} & \text{for } N-L+1 < s \leq N \end{cases} \quad (5)$$

where $k = 1, \dots, m$. The time series is then reconstructed by adding the time series structures together by simple summation, i.e. $\tilde{y}_t = \sum_{k=1}^m \tilde{y}_t^{(k)}$ for $t = 1, \dots, N$. Optimality of the Hankelization (diagonal averaging) operation can be studied in Chapter 6 of Golyandina *et al.* (2001).

2.2. Separability

The “separability” of elementary SSA components can be used to assist with the grouping stage in SSA and is also useful in assessing how separable elementary time series structures in the original series is when applying SSA decomposition. “Separability” of elementary SSA components is measured in terms of weighted correlations (w-correlations) of the elementary time series structures produced by the SSA decomposition and reconstruction stages without grouping being applied. The separability of time series structures formed using the grouping stage can, however, also be used in assessing separability of time series structures so formed. We will only consider here the separability of elementary time series structures in SSA decomposition. Let us assume here that two elementary time series structures $\{\tilde{y}_t^{(1)}\}_{t=1}^N$ and $\{\tilde{y}_t^{(2)}\}_{t=1}^N$ were produced after a time series was embedded, SVD applied and that the latter two series were reconstructed using only a single eigen-triple in their respective reconstructions by diagonal averaging. Golyandina *et al.* (2001) stipulates that the weights applied in calculating the w-correlations are given by

$$w_i = \begin{cases} i & \text{for } 1 \leq i \leq L^* \\ L^* & \text{for } L^* + 1 \leq i \leq n^* \\ N - i + 1 & \text{for } n^* + 1 \leq i \leq N \end{cases} \quad (6)$$

where $n = N - L + 1$, $L^* = \min(L, n)$, $n^* = \max(L, n)$ and the w-correlations are calculated using

$$\rho_{12}^{(w)} = \frac{\sum_{i=1}^N w_i \tilde{y}_i^{(1)} \tilde{y}_i^{(2)}}{\sqrt{\sum_{i=1}^N w_i \tilde{y}_i^{(1)} \tilde{y}_i^{(1)}} \sqrt{\sum_{i=1}^N w_i \tilde{y}_i^{(2)} \tilde{y}_i^{(2)}}}. \quad (7)$$

Small w-correlations are an indication that elementary time series structures are nearly w-orthogonal and therefore separable. A natural graphical representation of the w-correlations is to produce a matrix consisting of the absolute values of the w-correlations and to graphically plot the correlations using a contiguous 20 interval gray color scaling which represents correlations in the range $[0, 1]$. An example is considered in Section 4 of this article. Large adjacent w-correlations in the matrix, based on elementary time series structure reconstructed using eigentriples $(\sqrt{\lambda_i}, \mathbf{u}_i, \mathbf{v}_i)$ for $i = 1, \dots, L$ where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_L > 0$, would indicate that consecutive elementary time series structures are not separable and should rather be grouped. The latter is also an indication that time series structures are mixed and are useful in identifying, if encountered, which harmonics are possibly interfering with each other or trend components in the decomposition stage. Hence, the w-correlation matrix can essentially assist in assuring correct groupings of elementary matrices in the grouping stage. The w-correlations based on time series structures after grouping can also assist in evaluating the separability of time series structure resulting from the grouping stage in SSA.

It will be shown as part of an example (cf. Section 4) that the w-correlations can also assist to decide whether a log-transformation is necessary to assist SSA decomposition. Log-transformations are generally not required in SSA, as the methodology can handle amplitude modulated seasonal periods.

3. Forecasting

With the time series structure identified as described above, it is natural that forecasts would be required using SSA. The earliest reference of forecasting using SSA dates back to Danilov (1997). Venter (1998) showed that a one-period-ahead forecast can be obtained using

$$\hat{y}_{N+1} = \frac{\langle \mathbf{P}_{L_r} \mathbf{x}_{n+1}^{(0,L)}, \mathbf{e}_L \rangle}{1 - \|\mathbf{P}_{L_r} \mathbf{e}_L\|^2} \quad (8)$$

where $\langle \mathbf{P}_{L_r} \mathbf{x}_{n+1}^{(0,L)}, \mathbf{e}_L \rangle$ denotes the vector inner product, $\mathbf{P}_{L_r}$ is the projection matrix formed using the leading r left singular vectors $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ obtained during SVD of the trajectory matrix, $\mathbf{e}_L$ is column vector of the identity matrix $\mathbf{I}_L$ and $\mathbf{x}_{n+1}^{(0,L)} = [y_{N-L+1} \ \cdots \ y_N \ 0]^T$. Recurrent one-period-ahead forecasts can be obtained by reusing previous one-period-ahead forecasts and replacing them into $\mathbf{x}_{n+2}^{(0,L)}$. The latter translates into a re-embedding of the time series with concatenated one-period-ahead forecasts into a new trajectory matrix. The recurrent one-period-ahead forecast for time period $N+2$ can then, for example, be constructed using $\mathbf{x}_{n+2}^{(0,L)} = [y_{N-L+2} \ \cdots \ y_N \ \hat{y}_{N+1} \ 0]^T$ in the above formulation. It is not difficult to show (De Klerk, 2002) that the above formulation of a one-period-ahead forecast is similar to formulae presented in Golyandina et al. (2001). It is also

clear that a one-period-ahead forecast exists if and only if $\|\mathbf{P}_{L_r}\mathbf{e}_L\| \neq 1$, which is called the verticality condition in Golyandina et al. (2001).

There are a number of variants when applying forecasting using SSA. Golyandina et al. (2001) proposes that the reconstructed series $\{\tilde{y}_t\}_{t=1}^N$ can be used in producing forecasts, i.e. that $\mathbf{x}_{n+1}^{(0,L)} = [\tilde{y}_{N-L+1} \ \cdots \ \tilde{y}_N \ 0]^T$. Alternatively the original time series can be used in (8) to produce forecasts.

4. Application

SSA methodology will now be employed to identify time series structure in a South African tourism series. For this purpose we will consider series T110444 which was obtained from Statistics South Africa. This series consists of the monthly foreign travellers from Switzerland to South Africa. We will only consider the period June 1998 to April 2013. It is evident from a time series plot (cf. Figure 2, below) that the time series exhibits trend, monthly seasonality and a few other harmonic oscillatory patterns. The Time Series Forecasting System (TSFS) module of SAS version 9.3 software was also used to automatically fit appropriate time series models and produce forecasts. We will compare forecasts produced by seasonal exponential smoothing (SES) and SARIMA models by those produced by SSA recurrent one-period-ahead forecasting.

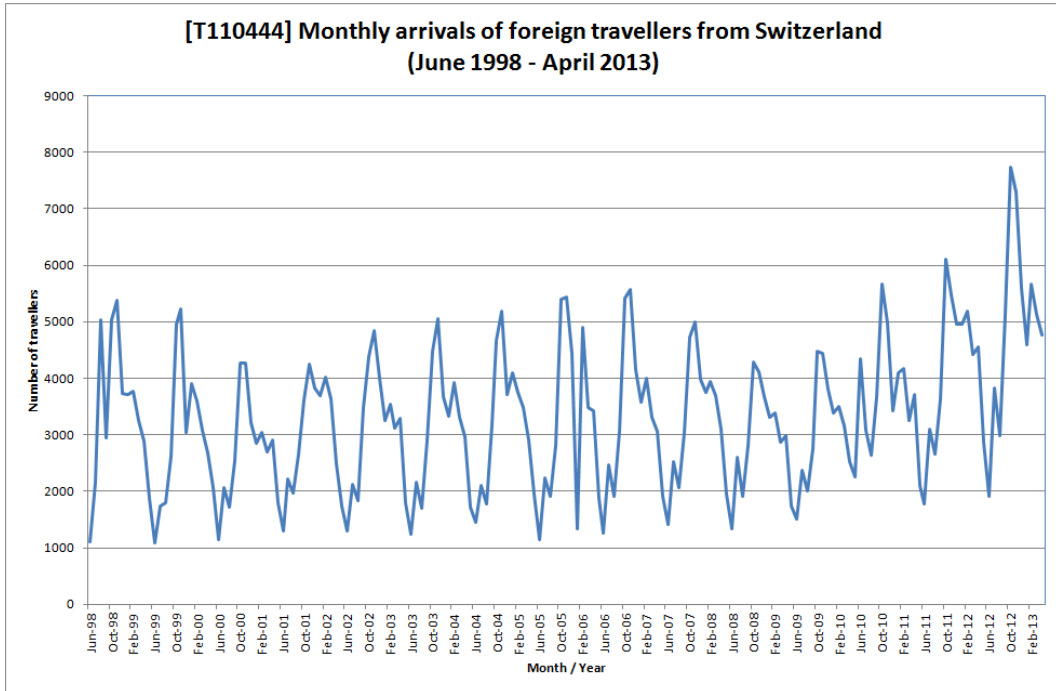


Figure 2: Monthly arrivals of foreign travellers to South Africa from Switzerland [June1998 - April2013].

4.1. SSA methodology

The tourism time series was split into a training set (June 1998 to October 2012) and a set for out-of-sample forecasting comparisons (November 2012 to April 2013). Golyandina et al. (2001) recommends that the window length used in embedding a time series be set equal to some integer multiple of the periodicity of the largest seasonal period in the time series, which in this case is 12 (monthly periodicity). The training series length is $N = 173$ and maximum choice of window length equal to $\text{floor}[(N + 1)/2] = 87$. The choice of window length was set to $L = 84$.

SSA methodology (embedding followed by singular value decomposition) was then applied to both the non-transformed and log-transformed training sets. Studying plots of the individual singular vectors based on the non-transformed set (cf. Figure 3) and paired eigenvectors (cf. Figure 5) reveals that the grouping [1], [2,3], [4,5], [6,7], [8,9], [10], [11,12] can be used to explain [trend], [seasonality with period 12], [seasonality with period 6], [seasonality with period 4], [seasonality with period 3], [trend] and [seasonality with period 2.4]. Individual and paired plots of singular vectors (cf. Figures 4 and 6) extracted from the log-transformed set reveal that the grouping [1], [2,3], [4,5], [6,7], [8,9], [10,11] and [12] can be used to explain [trend], [seasonality with period 12], [seasonality with period 6], [seasonality with period 4], [seasonality with period 2.4], [seasonality with period 3] and [trend].

Inspection of the w-correlation matrices based on the non-transformed training set's elementary components (cf. Figure 9) and log-transformed set's elementary components (cf. Figure 10), reveals time series separability issues for the non-transformed series. It is clear from Figure 9 that the groupings [1], [2,3], [4,5] and [6,7] should be used due to large w-correlations in adjacent singular vectors and that there does not seem to be any other separability issues for the latter groupings. Starting from singular vector [8], there seems to be separability issues for singular vectors [8-10]. Based on the plot of individual singular vectors in Figure 3, it was clear that the trend component in singular vector [10] was contaminated by the seasonal period in singular vector grouping [8,9]. The problem is also identified in the periodogram based on singular vector [10] as graphically depicted in Figure 11. The reconstruction of elementary time series structure based on eigentriple [10] as depicted in Figure 7 also exhibits the seasonally contaminated trend pattern. When perusing through the w-correlation matrix representation in Figure 10, which is based on the log-transformed training set decomposition, it is evident that separability issues are only experienced beyond singular vector [12], which is not deemed a problem as the remaining individual and paired singular vector plots indicate noisy structures that are of no use in the time series decomposition. It suffices to conclude that the leading 12 eigentriples are seemingly representing the combined time series structures appropriately.

It should be clear from the latter discourse why the log-transformed training set was rather used in reconstruction of the combined time series structures and ensuing forecasting.

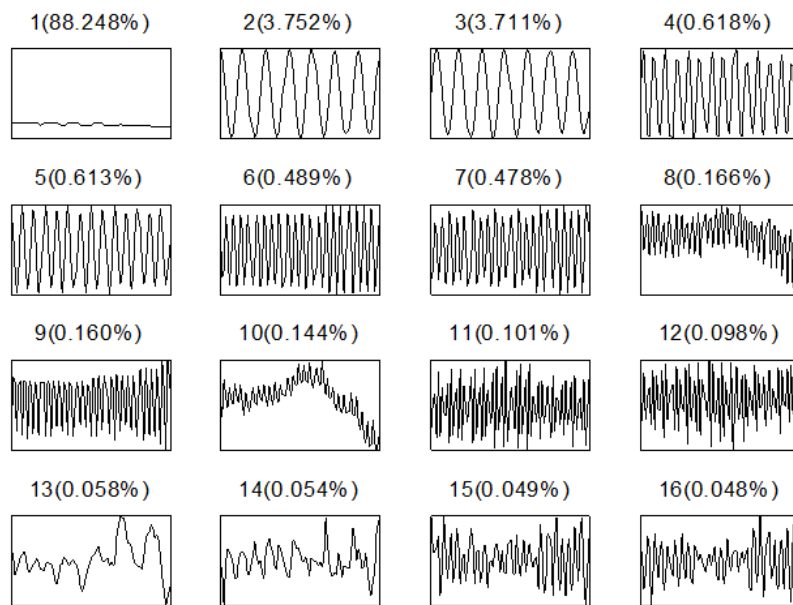


Figure 3: Plot of individual singular vectors (non-transformed time series).

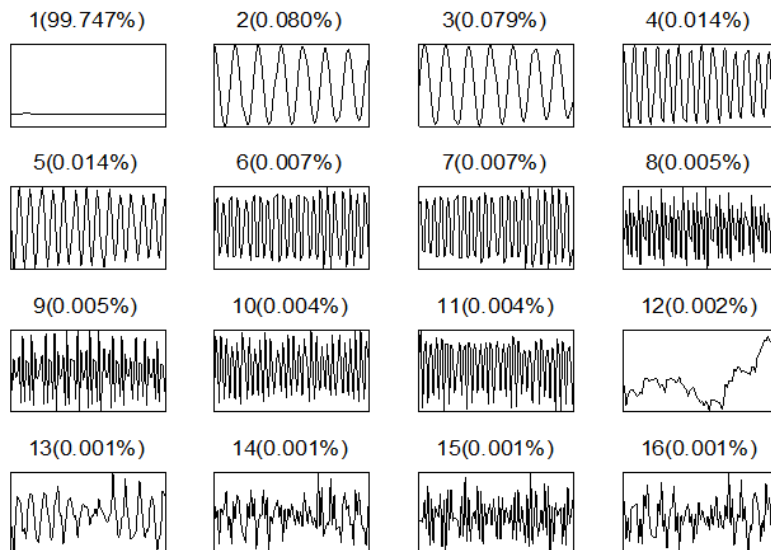


Figure 4: Plot of individual singular vectors (log-transformed time series).

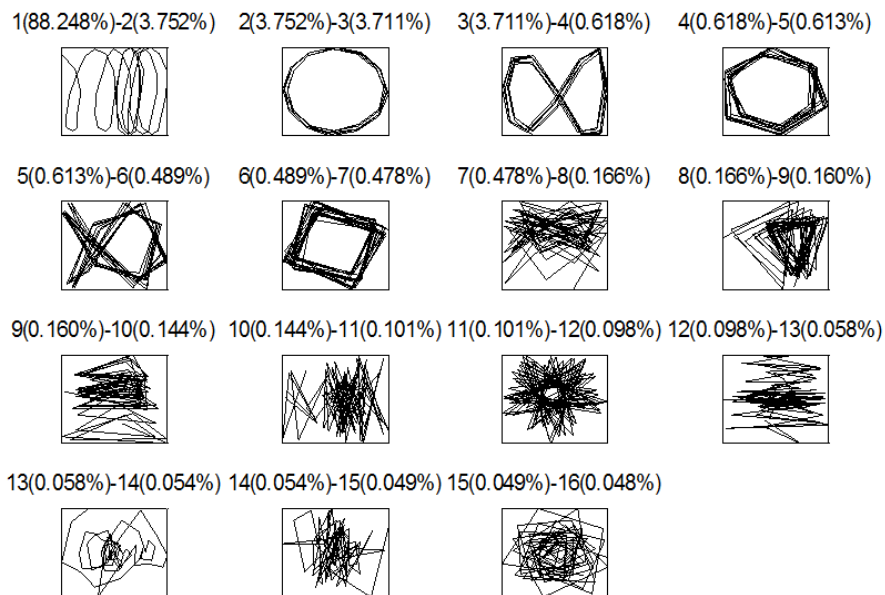


Figure 5: Plot of paired singular vectors (non-transformed series).

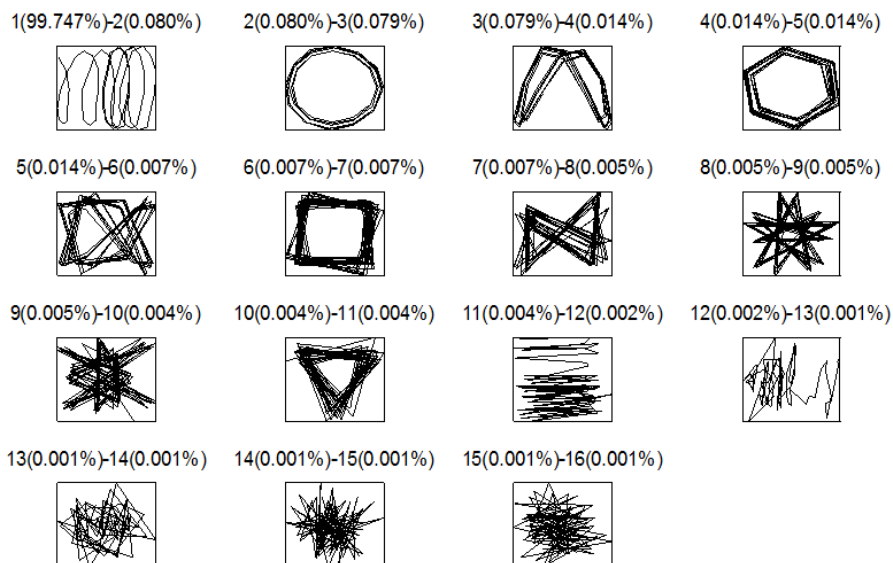


Figure 6: Plot of paired singular vectors (log-transformed series).

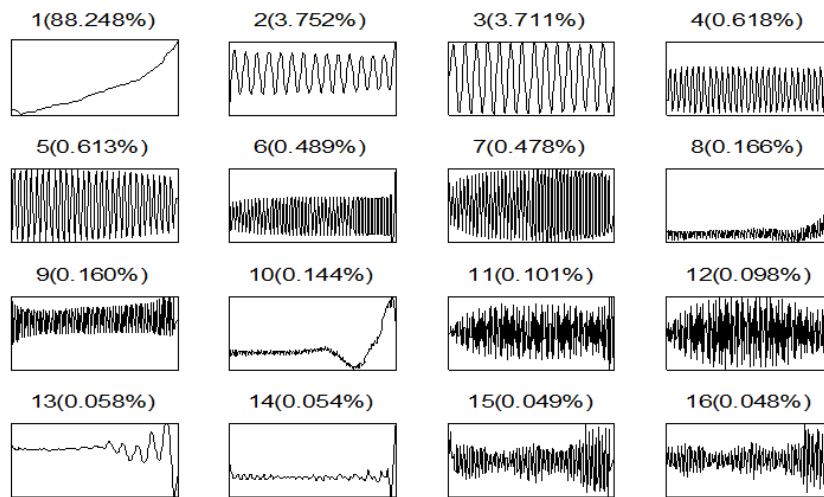


Figure 7: Plot of reconstructed elementary components (non-transformed series).

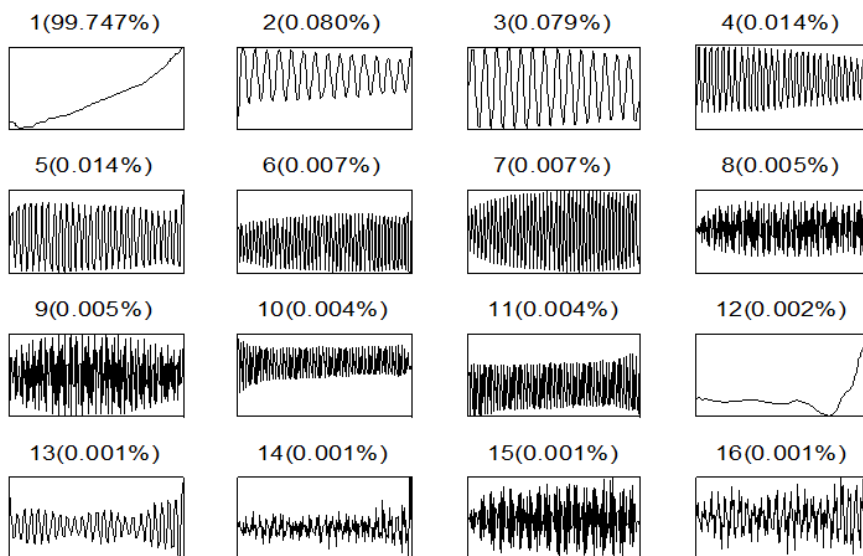


Figure 8: Plot of reconstructed elementary components (log-transformed series).

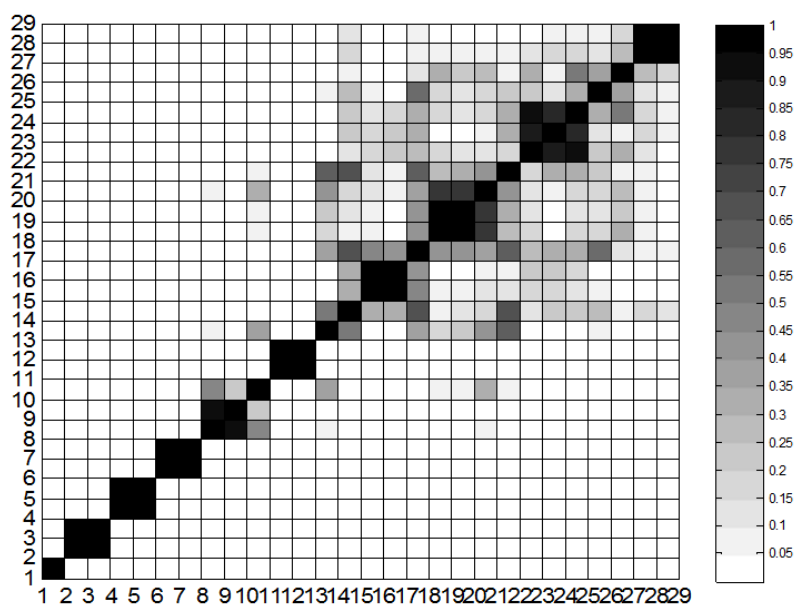


Figure 9: Plot of w-correlations based on elementary components (non-transformed series).

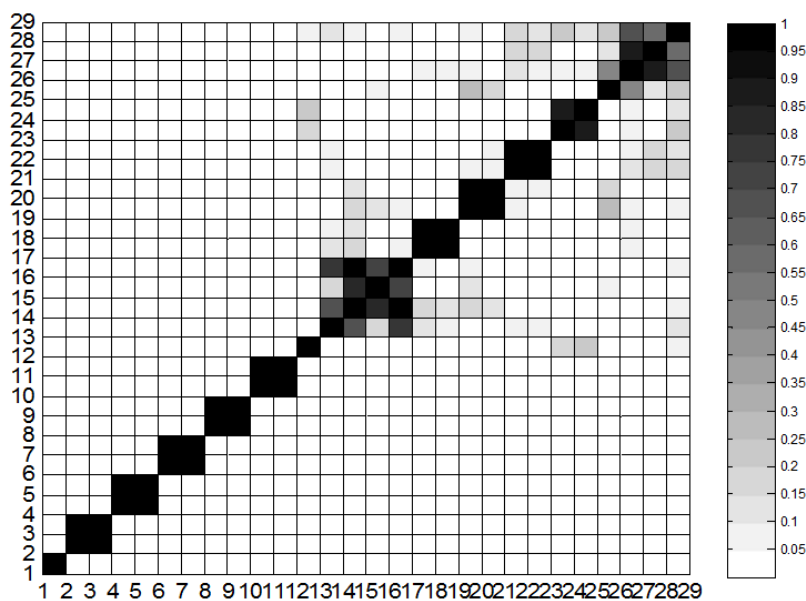


Figure 10: Plot of w-correlations based on elementary components (log-transformed series).

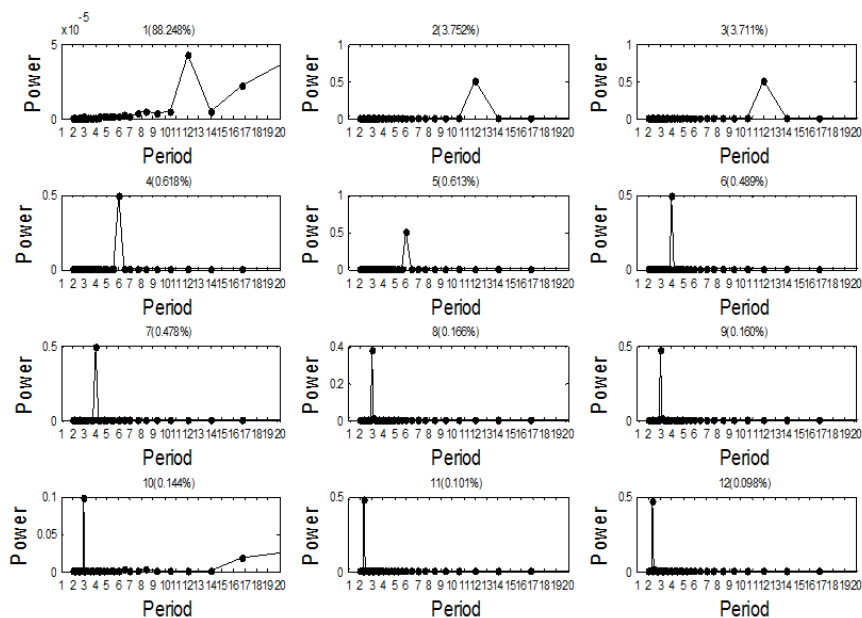


Figure 11: Periodograms based on singular vectors (non-transformed series).

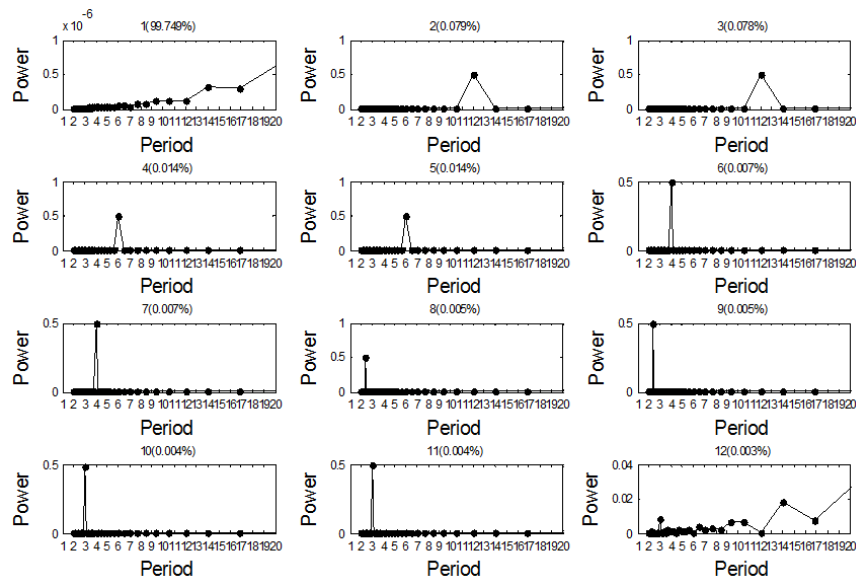


Figure 12: Periodograms based on singular vectors (log-transformed series).

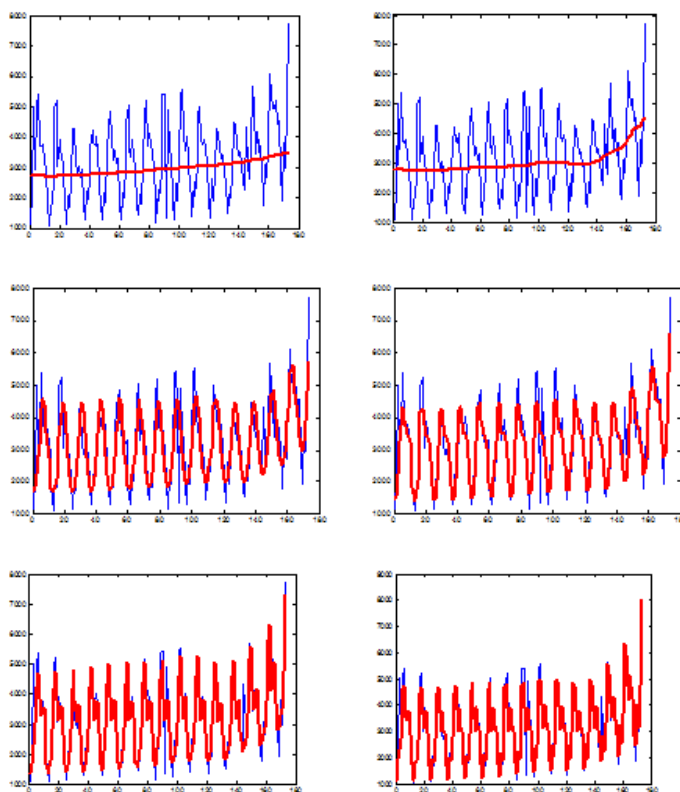


Figure 13: Reconstructed series using eigentriple groupings [(1); (1,12); (1:3,12); (1:5,12); (1:7,12); (1-12)] (left to right, top to bottom).

Figure 13, above, illustrates reconstruction of the training set (June 1998 – October 2012) using the eigentriple groupings (1), (1,12), (1:3,12), (1:5,12), (1:7,12) and the leading 12 eigentriples extracted from the log-transformed training set and the reconstruction being back-transformed to the original scale. It is clear that the reconstruction based on the leading 12 eigentriples seem to capture the time series structure very well in that trend and all harmonics have been accounted for in the training set.

4.2. Time series models fitted using SAS/ETS software

The Time Series Forecasting System (TSFS), which forms part of the SAS/ETS module, was used to automatically fit time series models to the training set (June 1998 to October 2012). SAS version 9.3 software was used for this purpose. The package was used to find the optimal set of suitable time series models that would minimize the Root Mean Square Error (RMSE) over the training set. The models that were automatically evaluated by the TSFS module are ranked in Table 2 according to RMSE performance.

Table 2: List of models automatically evaluated by TSFS.

Model	RMSE
Log Seasonal Exponential Smoothing (SES)	505.65
Seasonal Exponential Smoothing	521.31
Seasonal Dummy	629.49
Log Seasonal Dummy	631.30
Log $ARIMA(0, 1, 1)_{12} NOINT$	665.16
$ARIMA(0, 1, 1)_{12} NOINT$	665.90
$ARIMA(2, 0, 0)(1, 0, 0)_{12}$	699.31
Log $ARIMA(2, 0, 0)(1, 0, 0)_{12}$	702.10

It is clear from the above table that the TSFS module evaluated models that are capable of modelling time series seasonality and that a log-transformation was also considered as part of the modelling approach. The best performing model was the seasonal exponential smoothing model applied to the log-transformed time series. It is interesting to note that the TSFS module in automated mode did not consider the well-known Holt-Winters smoothing methods (additive or multiplicative), which is capable of handling series that contain seasonality and trend. The exponential smoothing model does not cater for trend in time series, but only a constant level. Hence, it seems that the trend does not play a major role according to the models selected by the TSFS module.

The development of exponential smoothing models is accredited to Brown (1959), Brown (1962) and Holt, Modigliani, Muth, Simon and et al. (1960) and the model has the following components, viz.

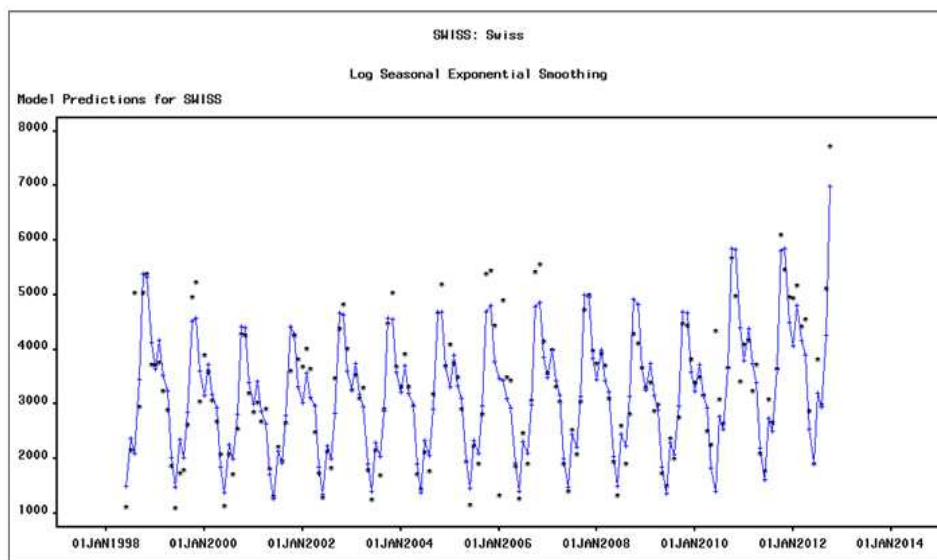
$$\begin{aligned}
 \text{Level : } L_t &= \alpha(Y_t - S_{t-s}) + (1 - \alpha)L_{t-1} \\
 \text{Seasonal : } S_t &= \gamma(Y_t - L_t) + (1 - \gamma)S_{t-s} \\
 \text{Forecast : } F_{t+m} &= L_t + S_{t-s+m}
 \end{aligned}$$

where α and γ are respectively the level and seasonal smoothing parameters and the seasonal length is denoted by s . The number of periods being forecasted ahead is denoted by m . Parameter estimates produced by the TSFS module for the exponential smoothing model applied to the log-transformed training set are summarized in Table 3, below. The actual and fitted model is graphically depicted in Figure 14, below. It is clear from the reported test statistics in Table 3 that the LEVEL smoothing weight is statistically significantly different from zero ($p < 0.001$) and that the SEASONAL smoothing weight is not statistically significantly different from zero ($p = 0.9606$). The latter is not considered an issue, as without this weight the model would not make sense. The smoothing seasonal factors do not have statistical tests attached to them.

Table 3: Parameter estimates of Exponential Smoothing (fitted to log-transformed training set).

Model Parameter	Estimate	Standard Error	T	Prob> T
LEVEL smoothing weight	0.15211	0.0289	5.0994	<0.001
SEASONAL smoothing weight	0.00100	0.0202	0.0495	0.9606

Model Parameter	Estimate	Model Parameter	Estimate
Residual Variance (sigma squared)	0.02948	Smoothed Seasonal Factor 6	-0.73461
Smoothed Level	8.36303	Smoothed Seasonal Factor 7	-0.22597
Smoothed Seasonal Factor 1	0.12014	Smoothed Seasonal Factor 8	-0.33372
Smoothed Seasonal Factor 2	0.25508	Smoothed Seasonal Factor 9	0.02778
Smoothed Seasonal Factor 3	0.09607	Smoothed Seasonal Factor 10	0.49221
Smoothed Seasonal Factor 4	0.01870	Smoothed Seasonal Factor 11	0.48997
Smoothed Seasonal Factor 5	-0.43220	Smoothed Seasonal Factor 12	0.23292

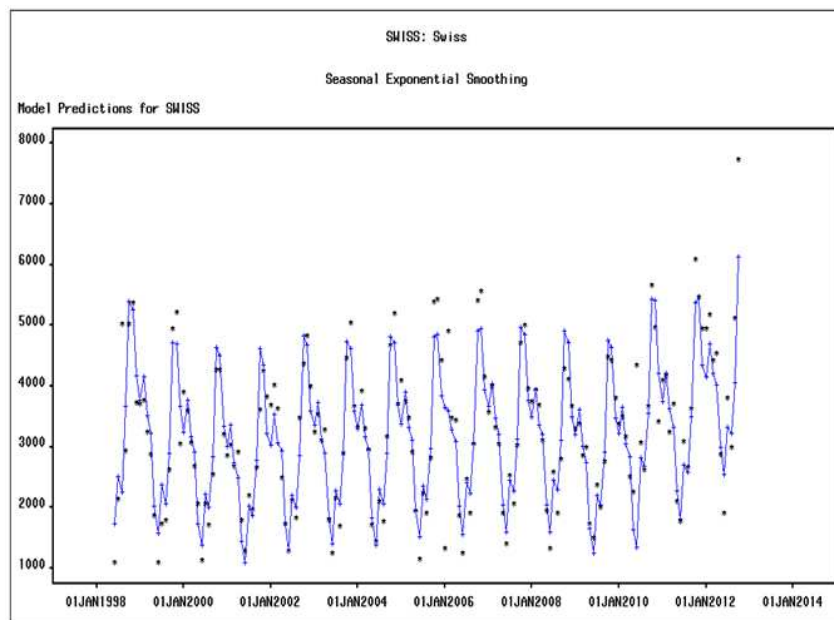
**Figure 14:** Time series versus fitted Log Seasonal Exponential Smoothing (solid line).

For sake of thoroughness the parameter estimates of an exponential smoothing model fitted to the non-transformed test set is reported in Table 4 , below. Figure 15 exhibits the actual time series versus the fitted exponential smoothing.

Table 4: Parameter estimates of Exponential Smoothing (fitted to non-transformed training set).

Model Parameter	Estimate	Standard Error	T	Prob> T
LEVEL smoothing weight	0.19213	0.0347	5.5438	<0.001
SEASONAL smoothing weight	0.00100	0.0158	0.0631	0.9497

Model Parameter	Estimate	Model Parameter	Estimate
Residual Variance (sigma squared)	274947	Smoothed Seasonal Factor 6	-1699
Smoothed Level	4684	Smoothed Seasonal Factor 7	-803.21061
Smoothed Seasonal Factor 1	263.24272	Smoothed Seasonal Factor 8	-992.73426
Smoothed Seasonal Factor 2	654.61980	Smoothed Seasonal Factor 9	-104.26829
Smoothed Seasonal Factor 3	87.14021	Smoothed Seasonal Factor 10	1757
Smoothed Seasonal Factor 4	-145.28033	Smoothed Seasonal Factor 11	1689
Smoothed Seasonal Factor 5	-1279	Smoothed Seasonal Factor 12	578.23468

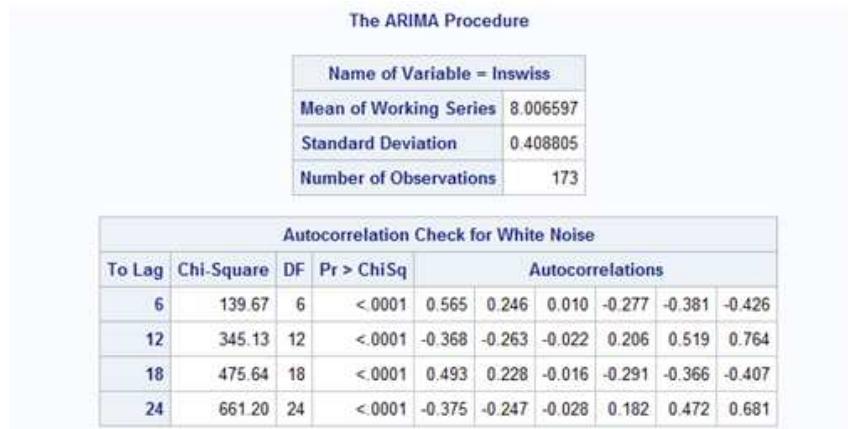
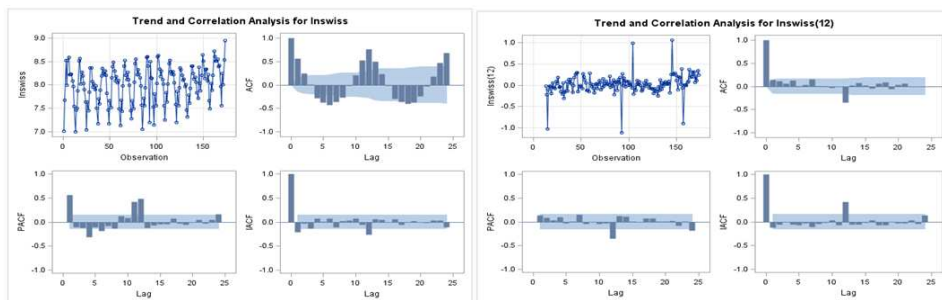
**Figure 15:** Time series versus fitted Seasonal Exponential Smoothing (solid line).

The TSFS module also fitted a $SARIMA(0, 1, 1)_{12}$ (with no intercept) model to the log-transformed training set of which parameter estimates are reported in Table 5, below. The fitted model is therefore of the form $(1 - B^{12})(1 - 0.54898B^{12})\ln(y_t) = z_t$ where B denotes the usual backshift operator and it is clear that a 12th order seasonal difference was applied. The estimated parameter is also statistically significantly different from zero ($p < 0.001$).

Table 5: Parameter estimates of Log $SARIMA(0, 1, 1)_{12}NOINT$.

Model Parameter	Estimate	Standard Error	T	Prob> T
Seasonal Moving Average, Lag 12	0.54898	0.0839	6.5457	<0.0001
Model variance (sigma squared)	0.04219			

Figures 16 and 17 respectively contain the autocorrelation check for white noise and ACF, PACF and inverse PACF based on the log-transformed test set and series resulting from a 12th-order seasonal difference to the log-transformed test set. The white noise test confirms that the series is not white noise and the ACF does not support need for a difference to remove trend. Figure 18 depicts the actual versus fitted SARIMA model.

**Figure 16:** Autocorrelation check for white noise (log-transformed series).**Figure 17:** Autocorrelation check for white noise (log-transformed series).

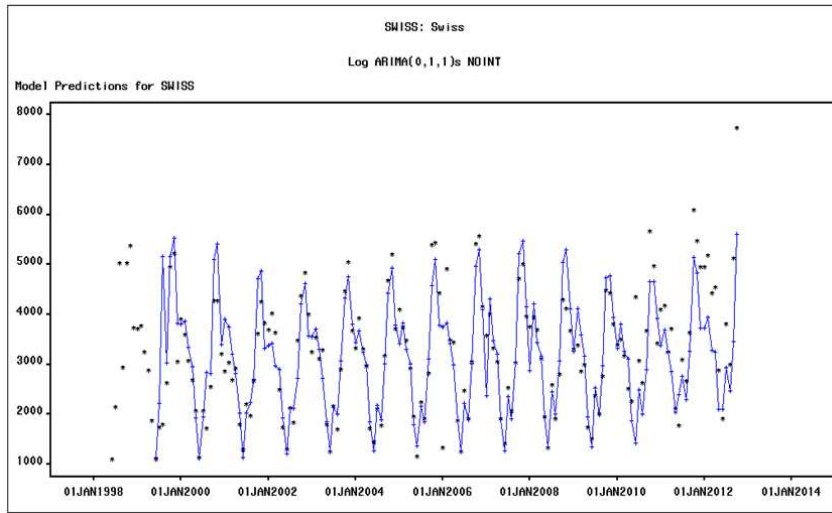


Figure 18: Autocorrelation check for white noise (log-transformed series).

4.3. Forecasting results of models

Forecasts were next produced for the period November 2012 – April 2013 using the top five models fitted by the TSFS module.

Table 6: Comparison of forecasting results for several time series methods (ranked by MAPE).

Model	Nov'12	Dec'12	Jan'13	Feb'13	Mar'13	Apr'13	MAE	MAPE
Original Data	7291	5597	4587	5671	5135	4761		
Log SES	7099	5492	4908	5594	4794	4439	226.3	4.38%
Log SSA ($L = 84, r = 12$)	7096	5604	5034	6118	5124	5041	231.2	4.42%
SSA ($L = 84, r = 12$)	6302	5273	5282	5508	5029	4946	410.3	7.22%
Simple Exponential Smoothing	6373	5262	4947	5338	4771	4538	422.2	7.34%
Log ARIMA (0, 1, 1) ₁₂ NOINT	5164	4278	4272	4500	3790	3817	1203.5	21.05%
Log Seasonal Dummy	5022	3884	3470	3954	3388	3136	1698.0	30.75%
Seasonal Dummy	4941	3830	3515	3907	3340	3108	1733.5	31.33%

Forecasts were also produced using SSA methodology where both the log-transformed and non-transformed test set was used to produce an SSA “model”. A window length of $L = 84$ and the leading $r = 12$ eigentriples were used to apply SSA methodology to produce recurrent one-period-ahead forecasting as described in this paper. Note that forecasts were back transformed, where applicable, to produce the measures of forecasting accuracy, i.e. MAE (mean absolute error) and MAPE (mean absolute percentage error) as summarized in Table 6, above. It is clear from the

results that the SSA method compares very favourably with existing time series methods.

The forecasting performance of SSA illustrated here compared to traditional time series methods supports earlier evidence by Hassani (2007) that SSA forecasting is more accurate than several well-known time series forecasting methods.

5. Concluding remarks

It has been demonstrated how SSA can be used to identify time series structure in a South African tourism time series. The method is powerful and has a very appealing non-parametric approach. It was also clear from forecasting results that forecasting accuracy using SSA is on par with that of traditional time series methods such as seasonal exponential smoothing and Box-Jenkins ARIMA models. Code developed in MATLAB as part of this research is available from the author on request. The MATLAB Statistics Toolbox is, however, required to operate the code successfully. An Rssa package developed in R <http://cran.r-project.org/web/packages/Rssa/> by researchers at St. Petersburg State University (Russia) is available and is capable of SSA analysis and forecasting. A detailed example of the application of SSA using the R package, which is due to Nina Golyandina and Anton Korobeynikov, can be obtained from link <http://arxiv.org/pdf/1206.6910v2.pdf>.

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